

Data Analysis - Zipline Systems Engineering

PART 1

FLIGHT METRICS AND INTERPRETATION

	RECOMMENDED METRIC	WHY IT MATTERS
1	Flight Success Rate	Measures the percentage of flights that successfully completed delivery. A declining success rate directly impacts customer satisfaction and operational efficiency.
2	Total Deliveries Completed	Measures business output and whether Zipline is meeting customer demand.
3	Average Delivery Delay (<i>Actual Delivery Time - Estimated Delivery Time</i>)	Indicates how reliably Zipline is in meeting promised delivery times. Even successful flights can hamper customer experience if consistently late.
4	Failure Reason Distribution	Identifies the biggest operational bottlenecks preventing successful deliveries. Enables prioritization of engineering and operational improvements.
5	Average Flight Duration	Helps monitor operational efficiency and detect routing, weather, or aircraft performance issues over time.

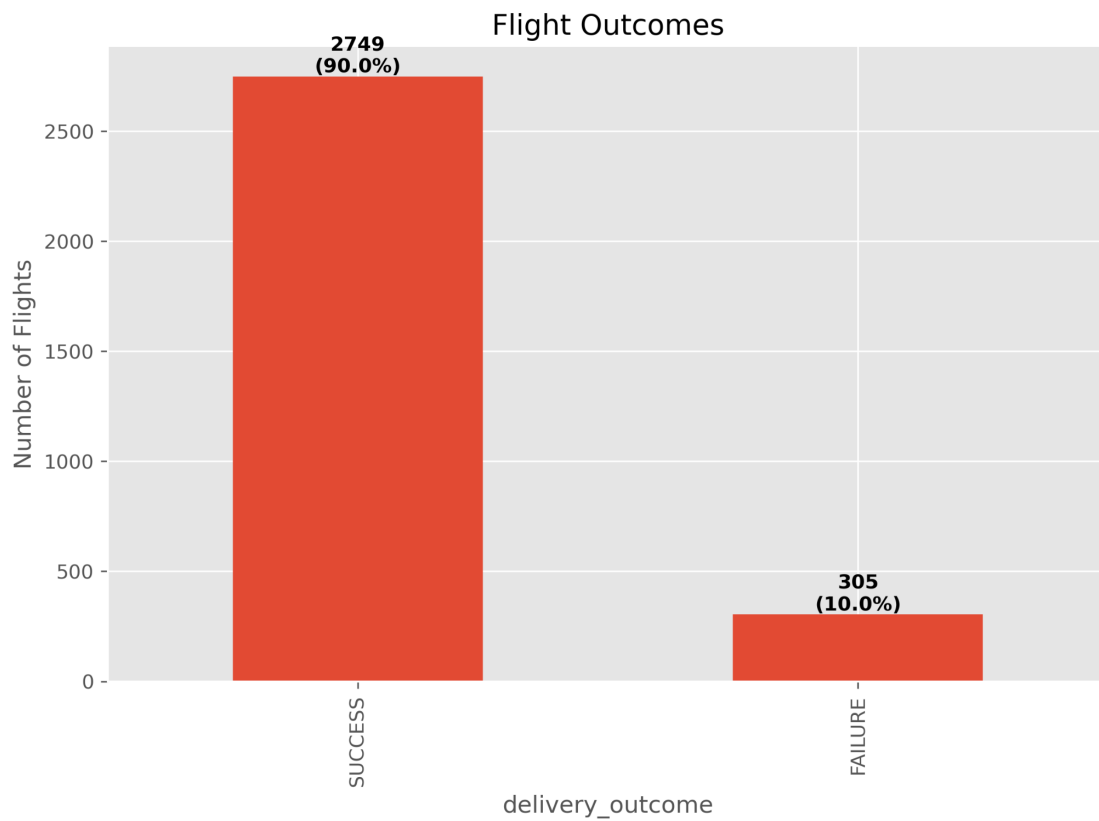
TOOLS AND METHODOLOGIES

I completed the data exploration, cleaning, analysis, and visualization in an offline Jupyter Notebook using Python. The notebook documents my entire analysis process. To run the notebook, please download both the **.ipynb** file and the accompanying **.xlsx** dataset from this [folder](#) and keep them in the same directory.

Every necessary file/ information is contained in this GITHUB repo

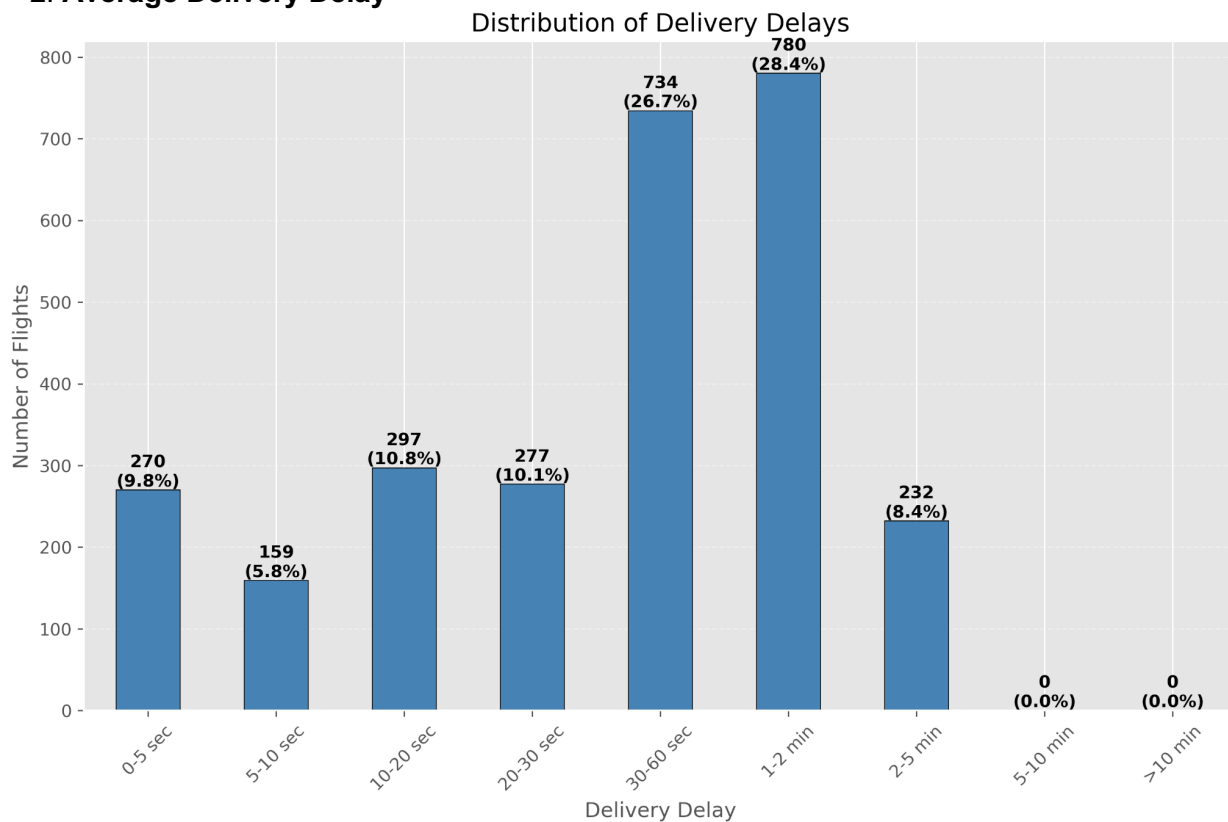
At the end of this part (1), I will include the data inconsistencies/ quality issues I observed and how I navigated through them.

1. Flight Success Rate & Total Deliveries Completed (Metrics 1 & 2)



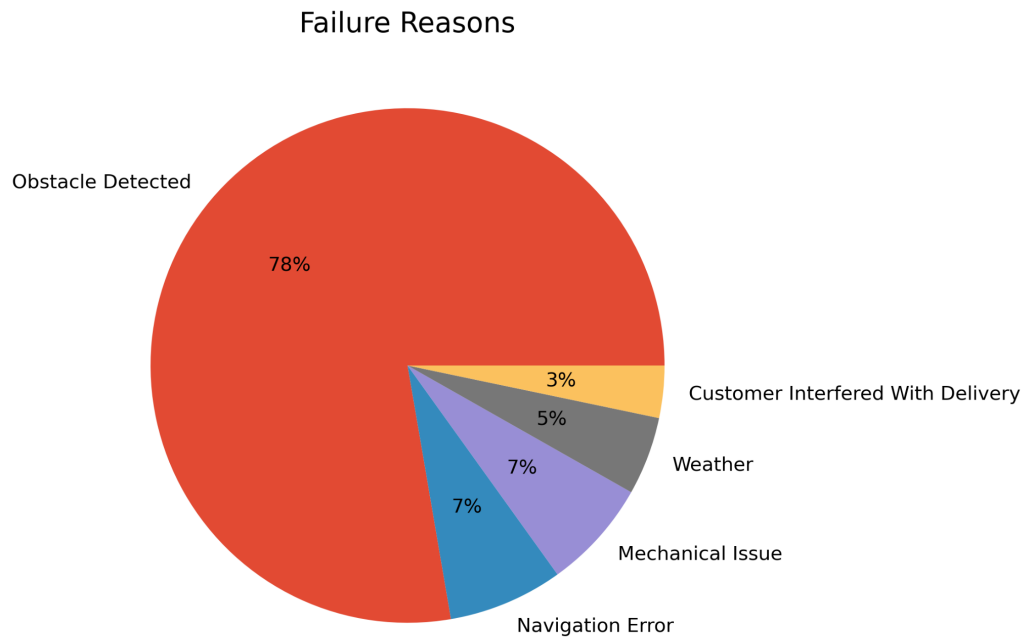
The chart above shows the proportion of successful flights alongside the total number of deliveries completed. Together, these metrics provide a high-level view of operational performance.

2. Average Delivery Delay

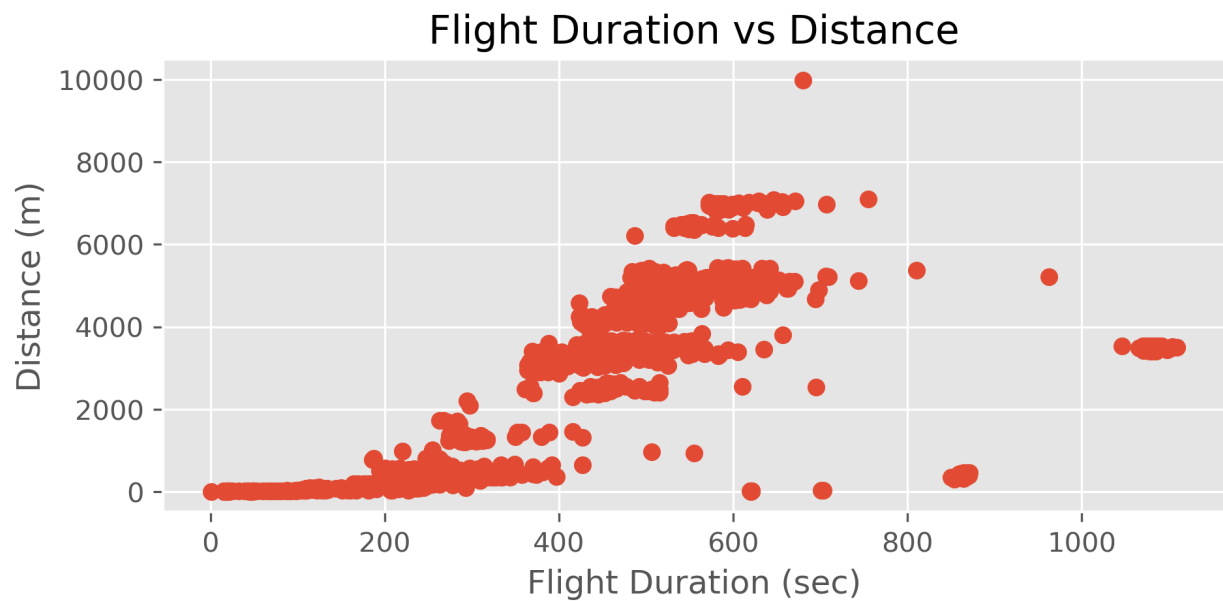


This *Distribution of Delivery Delays* chart measures the difference between actual and estimated delivery times, and then shows the distribution of the delays.

4. Failure Reason Distribution



5. Average Flight Duration



This chart tracks the average duration of completed flights over time. Significant increases or decreases may indicate changes in routing efficiency, weather conditions, aircraft performance, or operational procedures that require further investigation.

Executive Summary

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ZIPLINE FLIGHT DASHBOARD

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Total Flights: 3054

Successful Flights: 2749

Failed Flights: 305

Success Rate: 90.0%

Average Flight Duration: 389.9 sec

Average Delivery Delay: 52.8 sec

OPERATIONAL PERFORMANCE

Overall Flight Success Rate: 90% (9 of 10 flights):

The analysis shows that Zipline is performing at a **90.0% flight success rate** (2,749 successful deliveries out of 3,054 flights). While this looks good overall, it however means that about **one in ten** delivery attempts failed, which is likely below operational targets and represents lost deliveries and reduced customer satisfaction. The **10.0% failure rate** represents a meaningful opportunity to improve delivery volume and customer satisfaction.

EFFICIENCY

Average Flight Duration: 6 Minutes, 30 seconds (389.9s):

Flight duration generally increased with travel distance, indicating expected operational behavior without obvious efficiency outliers.

RELIABILITY

Average Delivery Delay: 52.8 seconds:

On an average, successful deliveries arrived approximately **one minute later than estimated**. While the delays are relatively small, they are consistent across nearly all successful flights, suggesting that estimated delivery times may be overly optimistic or operational buffers are insufficient.

Among successful deliveries, **55.3%** were completed between **30 seconds and 2 minutes later** than the estimated delivery time, with the largest group (**28.4%**) arriving between **1 and 2 minutes late**. Very few deliveries experienced delays exceeding two minutes, indicating that while deliveries are generally reliable, the ETA accuracy can be improved.

REASONS FOR FAILURES

Failure analysis reveals a clear operational bottleneck. **Obstacle Detected accounts for 78% of all failed deliveries**, making it by far the primary driver of unsuccessful flights. Navigation errors (7%), mechanical issues (7%), weather (5%), and customer interference (3%) contribute relatively little by comparison. Reducing obstacle-related failures would have the greatest impact on increasing successful deliveries and moving the organization closer to its flight success targets.

DATA INCONSISTENCIES AND CLEANING STEPS

These are the steps I took to improve the data quality while preserving as much of the original dataset as possible. Where possible, I corrected outliers/ wrong values by recalculating them from related fields rather than removing records. Only one record was dropped due to an extreme outlier that could not be reliably corrected.

Below is how I handled the different categories of issues:

1) Missing values

- a) I reviewed missing values across all columns using `df.isnull().sum()`
- b) Identified expected missing values, such as `actual_delivery_time_utc` for failed deliveries and `delivery_failure_reason` for successful deliveries.

2) Date and time fields (`takeoff_time_utc`, `estimated_delivery_time_utc`, `actual_delivery_time_utc`, and `docking_time_utc`) to datetime objects to enable time-based calculations.

3) Created new columns with derived metrics

- a) `delivery_delay_s` for successful deliveries to measure delivery delays.
- b) `success` as a boolean replica of `delivery_outcome`.

4) Handling Outliers

- a) Identified an outlier in `flight_duration_s`. I recalculated the value using `docking_time_utc - takeoff_time_utc` to ensure consistency with the flight timestamps.
- b) Identified an outlier in `docking_time_utc`. I recalculated the timestamp using `takeoff_time_utc + flight_duration_s` to correct the erroneous value.
- c) Identified one record with an unusually large `ground_based_distance_m` value. Excluded this record from the distance analysis to prevent it from disproportionately influencing the relationship between flight distance and duration.

5) Handled missing distance values for visualization

- a) Excluded records with missing `ground_based_distance_m` values from the distance versus flight duration visualization while retaining them in the main dataset.

PART 2

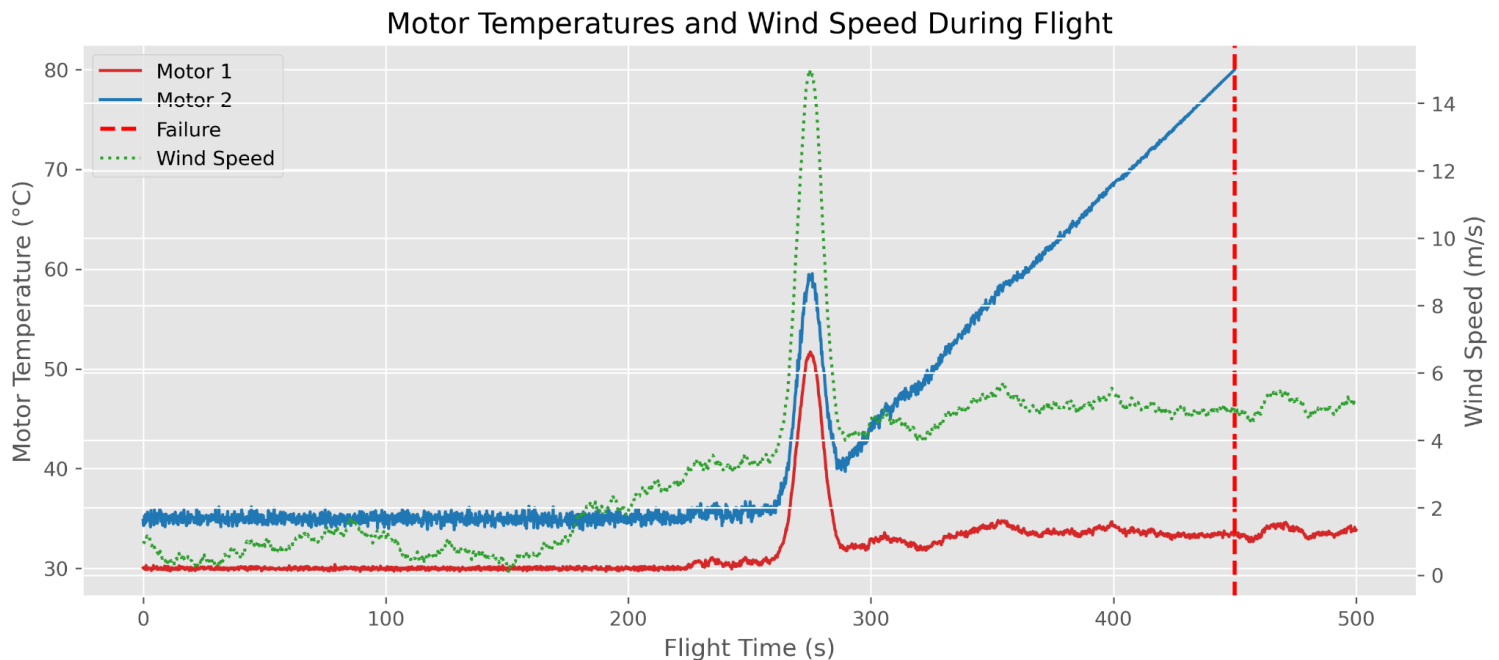
Flight Failure Analysis Summary

Aircraft: p2-zip-xyz

Flight Duration: 500.0 s

Failure Time: Approximately 450.0 s

Failure Alarm: FAILURE_MOTOR_2_FAILED



According to the data, the failure of Motor 2 is flagged at about **450 seconds**, but initial analysis shows that the abnormal behavior begins around **263 to 270 seconds** into the flight. This means the failure alarm was **reactive** and the final alarm was likely triggered after Motor 2 exceeds an internal safety threshold rather than when the underlying fault first appears.

TIMELINE OF EVENTS

As shown in the graph,

From 0 to about 260 s, both motors operated under stable conditions. Motor 1 remains around **30°C** and Motor 2 remains around **35°C**. The 5°C difference is consistent and likely normal.

At approximately 270 s, both motors experienced a sharp increase in temperature, coinciding with a significant increase in wind speed from approximately **4 m/s to 15 m/s** with Motor 1 peaking at around **52°C** and Motor 2 peaking at **60°C**.

After the wind speed returned to a stable level, the motors behaved very differently.

Motor 1 cooled rapidly and stabilized between **32°C and 34°C**, indicating that it recovered normally once the external load decreased.

Motor 2, however, failed to recover. Instead, its temperature continued to increase almost linearly, rising from approximately **40°C** after the wind event to over **80°C** by the time the failure alarm was triggered. This sustained and progressive increase is the strongest indicator of an underlying fault.

The data available thus suggests the high wind event was likely a trigger, not the root cause.

My assumed sequence of events is:

1. The aircraft encounters a strong wind gust.
2. Both motors work harder and heat up.
3. Motor 1 withstands the increased load and returns to normal.
4. Motor 2 sustains damage or develops an internal fault during the high-load event.
5. Even after wind speed returns to normal, Motor 2 continues generating excess heat.
6. The flight controller eventually detects the persistent abnormal condition and triggers the failure alarm.
7. Following the failure alarm, no additional Motor 2 temperature data are recorded. My assumption is that the motor or its associated telemetry shut down (although the available data do not allow the exact cause of the missing readings to be confirmed)